

WORMLOCKER 2.0 RANSOMWARE

Malware behavioral analysis project

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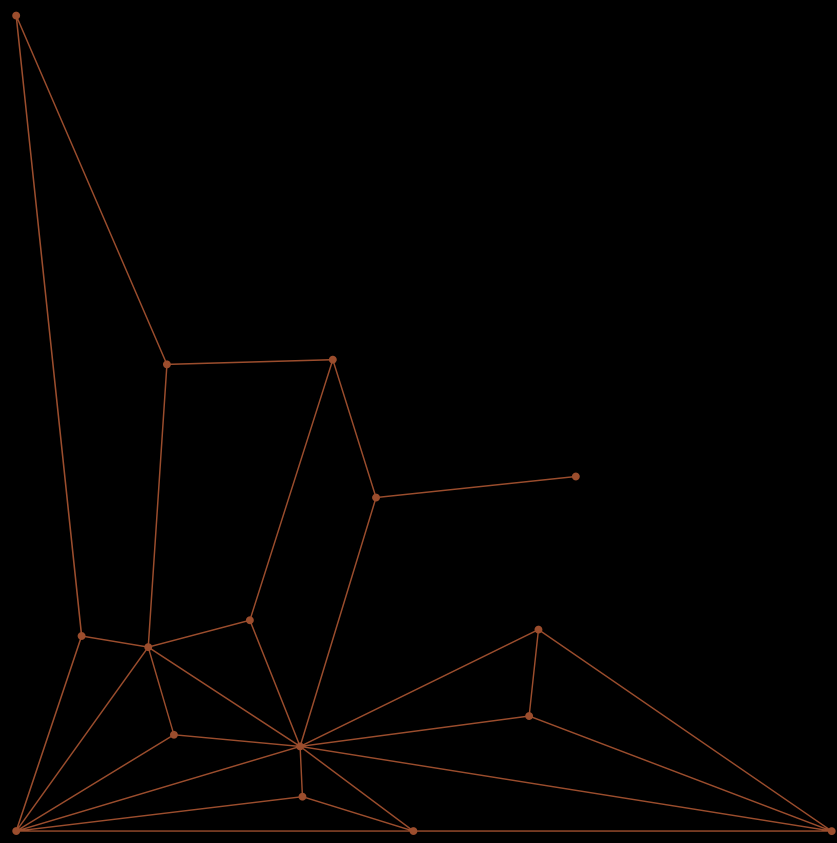
Attack
Simulation

Digital
Forensics

Static Analysis

Dynamic Analysis
& Monitoring

Conclusion



RANSOMWARE BEHAVIOURAL ANALYSIS DEFINITION

“Malware behavioral analysis is the process of observing a suspicious file or program in a controlled environment to see its actions.
We monitor things like changes to files, registry, network communication, processes, and system resources.”

THE RANSOMWARE WORMLOCKER 2.0: A REAL- WORLD CASE STUDY

- **Discovery & Target:**
- **Discovered:**
 - December 2020
 - Primary Target: European individual users
- **The Lure & Distribution:**
 - Distribution Method:** Disguised as a legitimate video converter software.
 - Social Engineering:** Tricks users into installing it, believing they are getting a useful tool.
- **The Ransom Demand:**
- **Ransom:** A relatively low 50 euros
- **Payment Method:** Paysafecard (a pre-paid voucher, hard to trace)

OUR CONTROLLED LAB ENVIRONMENT

Three-VM Architecture: [ATTACKER VM] ↔ [VICTIM VM] ↔ [MONITORING VM]



Protection

Fully Isolated Sandbox
Network: No risk to real
systems



Attacker VM (Kali Linux)

Where we host and control
the malware.



Victim VM

A clean Windows system for
infection.



Monitoring Server

kali



ATTACK SIMULATION

Drive-By Download Attack

- Victim visits a malicious webpage.
- The user downloads and executes the malicious software .
- In our simulation, WormLocker 2.0 locked the screen and displayed a ransom note.



PHASE 1: **BEHAVIOURAL AND FORENSIC ANALYSIS**

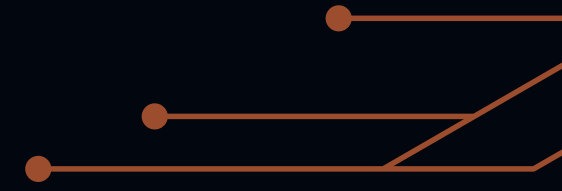
"Investigating the Compromised machine"



PHASE 2: STATIC ANALYSIS

"Understanding Without Execution"

PHASE 2: STATIC ANALYSIS



Step 1: Creating Unique File Fingerprints

PHASE 2: STATIC ANALYSIS



Step 1: Creating Unique File Fingerprints

Results:

MD5: 1b4f89bdb12a349de92ca7f1261e67a0

SHA1: f368916850332757d7ed2f0ee335c16b9c9fc95b

SHA256: d4c83205cf6f3098ab6a757312525f4d14a57a819306eeea...

PHASE 2: STATIC ANALYSIS

Step 2: Extracting Embedded Strings "Finding Hidden Intelligence"

1. **ASCII Strings (Basic Text)** : `strings -n 8 WormLocker.exe > strings_ascii.txt`
2. **Unicode Strings (Windows Text)** : `strings -n 8 -el WormLocker.exe > strings_unicode.txt`
3. **FLOSS (Obfuscated/Hidden Strings)**: `floss WormLocker.exe > floss_output.txt`

PHASE 2: STATIC ANALYSIS

PRIVILEGE ESCALATION

- /k takeown /f C:\Windows\System32 && icacLS C:\Windows\System32 /grant "%username%:F"
- Forced Shutdown: shutdown /r /t 0

DROPPED FILES

- C:\Windows\System32\LogonUlinf.exe
- C:\Windows\System32\LogonUltrtrue.exe
- C:\Windows\System32\ransom_voice.vbs

REGISTRY KEY

- DisableTaskMgr = 1
- SOFTWARE\...\Winlogon modifications

CRYPTOGRAPHY

- RijndaelManaged (AES encryption)
- EncryptFile / DecryptFile

Attacker's contact :

- wormlocker789@gmail.com
- Ransom: 50 euros via Paysafecard

DEVELOPER's TRACES

- C:\Users\Worm\source\repos\...
- Developer alias: "Worm"
- Debug build (not release)

.NET FRAMEWORK

- System.Windows.Forms (GUI)
- System.Drawing (graphics)
- Framework version: v4.0.30319

COMPONENT STRUCTURE

- Main Ransomware: WormLocker2.0.exe
- Screen Locker: wormlogon.exe (replaces LogonUI)
- Dropper/Dispatcher: Automatic_converter_rff_to_mp4.exe
- Audio Component: ransom_voice.vbs

PHASE 2: STATIC ANALYSIS

Step 3: Examining PE File Structure

Command: `pescan WormLocker.exe`

PHASE 2: STATIC ANALYSIS

Step 4: Identifying Development Tools

Command: `die WormLocker.exe`

PHASE 2: STATIC ANALYSIS

Step 5:Packer Detection

PHASE 2: STATIC ANALYSIS

Investigation Steps

Tool/Command	Result	Interpretation
Entropy Analysis	7.33	High entropy - typically indicates packing/compression
UPX Test	NotPackedException: not packed by UPX	Not a UPX-packed file
YARA Signatures	IsNET_EXE ✓ IsWindowsGUI IsPacked HasDebugData ✓	Flagged as packed BUT has debug data (unusual for packed files)
YARA Packer Detection	No specific packer matched	No known packer signatures found
monodis	Full assembly structure visible	This is a .NET executable with full metadata intact!
DnSpy	Complete source code visible: - Encryption methods readable - Decryption routines exposed - All classes and methods clear	CONFIRMED: NOT PACKED!

PHASE 2: STATIC ANALYSIS

Step 5:Packer Detection

Conclusion : **NOT PACKED**

High entropy came from compressed resources (images, audio files)
embedded in the executable, NOT code packing!

PHASE 2: STATIC ANALYSIS

Final Step : Collect indicators of Compromise (IOCs)

File-Based IOCs

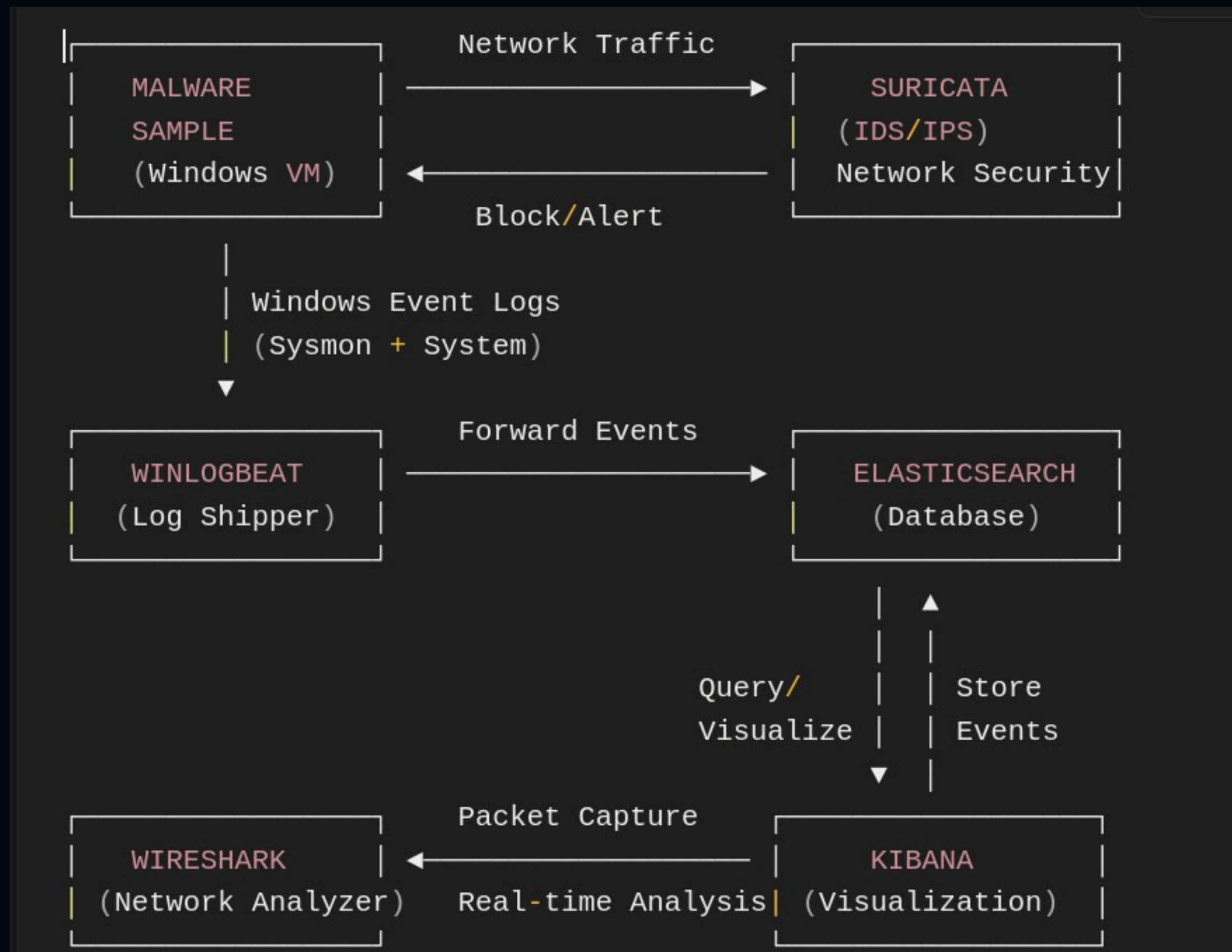
IOC Type	Value
SHA256 Hash	d4c83205cf6f3098ab6a757312525f4d14a57a819306eeea5c0d022b00b38cf3
MD5 Hash	1b4f89bdb12a349de92ca7f1261e67a0
SHA1 Hash	f368916850332757d7ed2f0ee335c16b9c9fc95b
Primary Filename	WormLocker2.0.exe
Fake Login	C:\Windows\System32\LogonUIinf.exe
Backup Login	C:\Windows\System32\LogonUItrue.exe
Audio Ransom	C:\Windows\System32\ransom_voice.vbs
Kernel Driver	%USERPROFILE%\Downloads\worm_tool.sys
Directory Marker	desktop.ini (modified)



PHASE 3: DYNAMIC ANALYSIS & MONITORING

"Execute and observe"

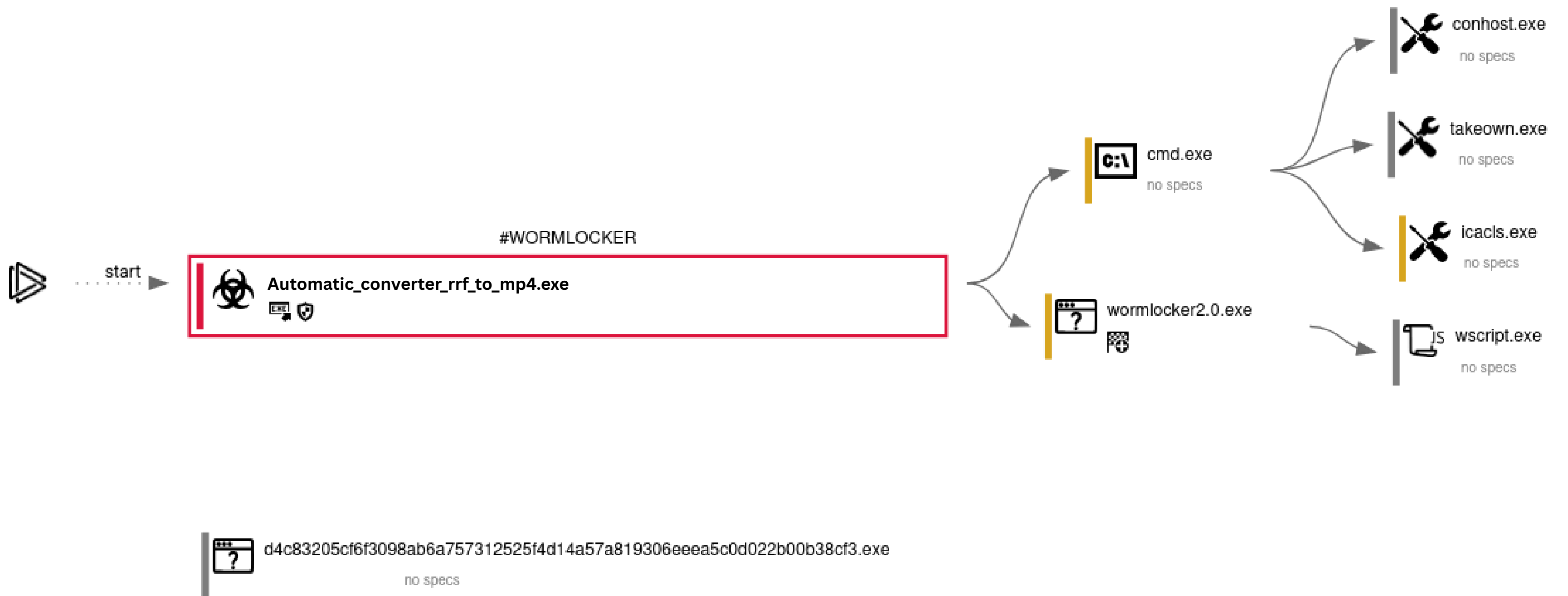
PHASE 3: DYNAMIC ANALYSIS & MONITORING





PHASE 4: **DECOMPILATION OF THE MALWARE**

"Decompile and Understand the code"



Conclusion